Free-fall lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Estimate g from measured motion and discuss measurement limits.

SUCCESS CRITERIA - I can...

1. Keep supplied examples separate from measurements.
2. Find acceleration from a velocity-time slope.
3. Use repeated trials and explain uncertainty.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “slope” and “trial” to explain your answer. Sentence starters are optional.

slope: vertical change divided by horizontal change on a graph

trial: one run or repetition of a test

acceleration: rate of change of velocity

01 Warm-up: bring back an earlier idea

1. Use the supplied gravity values: Gm_1m_2=**75 N·m²** and separation **5 m**.
2. **Find the force** and its direction.

Word help: separation = distance between the specified objects or centers

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $a = \Delta v/\Delta t$; $v_f = v_i + a\,\Delta t$.
Velocity-time slope estimates acceleration. Downward is positive. Use the approved apparatus; distinguish supplied from measured data.

Worked reasoning: Supplied trial values **8**, **10**, **12 m/s²** have mean **10 m/s²** and range **8**-**12**. The spread describes variation; averaging does not remove every systematic error.

Your turn: Plot the three values on a number line and **mark the mean**. **Name an error** that could shift all trials together.

Sentence starter: The step I would change is ... . I would change it because I found ... .

Trial	Estimated g (m/s²)
1	8
2	10
3	12

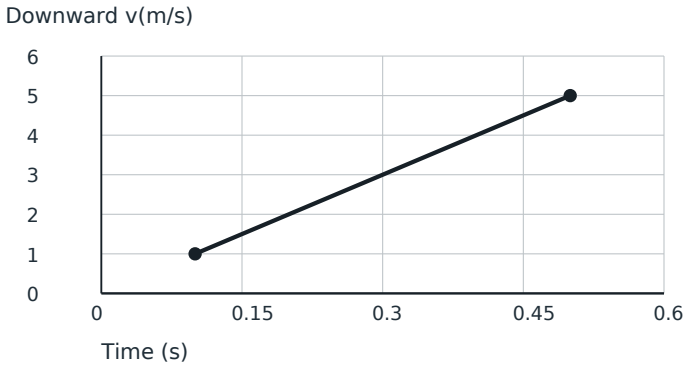
Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. Use the supplied velocity-time graph to **estimate acceleration**.
2. **Show the change in velocity** and elapsed time.

Word help: estimate = find an approximate value

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

Free-fall lab

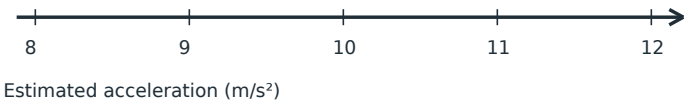
antocciphysics

Sentence starters are optional. You may plan in any language.

04 Apply the physics

Three video measurements of a falling ball give accelerations of **9**, **10** and **11 m/s²**. **Find the mean** and the lowest-to-highest interval. **Mark all three measurements** and the mean on the number line.

Sentence starter: I found … [unit]. This value means … in this situation.



Estimated acceleration (m/s²)

Marzano check - circle: ① ② ③

05 Explain your reasoning

Plan three repeated trials. **Name one part of the** setup you will keep the same, a measurement you will record and a reason not to include a badly pushed release without comment.

Sentence starter: My design includes … . It does what is needed by … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A box at rest has **18 N** right and **4 N** left. **Find the additional horizontal force** needed for balance and **show the sum**.
Draw the forces on the chosen object and **label their sources**.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.

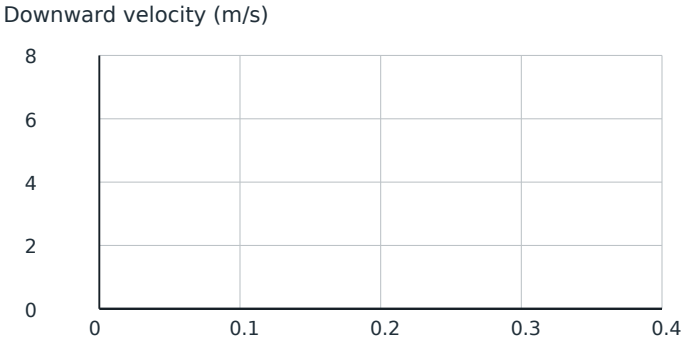
Add and label force arrows.

Marzano check - circle: ① ② ③

07 Exit check

A motion sensor records a falling ball speeding up from **4** to **6 m/s** over **0.4 s**. Take downward as positive.
1. **Draw the velocity-time segment** and **calculate its slope**.
2. **What acceleration** does this measurement imply?

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Two groups have values **9**, **10**, **11** and **5**, **10**, **15 m/s²**. **Compare means** and repeatability without declaring either perfectly accurate.

Sentence starter: Both … . They differ in … because … .

Group / trial	Values (m/s²)	Mean	Spread
1	____	____	____
2	____	____	____

Marzano check - circle: ① ② ③